

# DATABASE SYSTEMS

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## Chap 2~ DBMS DATA MODELLING

## 2. Objectives of the course

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- ✓ After completing this chapter, the reader should be familiar with the following concepts:
  - What is entity relationship model
  - Entity, Relationship and Attributes,
  - Entity classification
  - Attribute classification
  - Keys
  - E-R Diagram
  - Design of an E-R Database Schema
  - Reduction of an E-R Schema to Tables

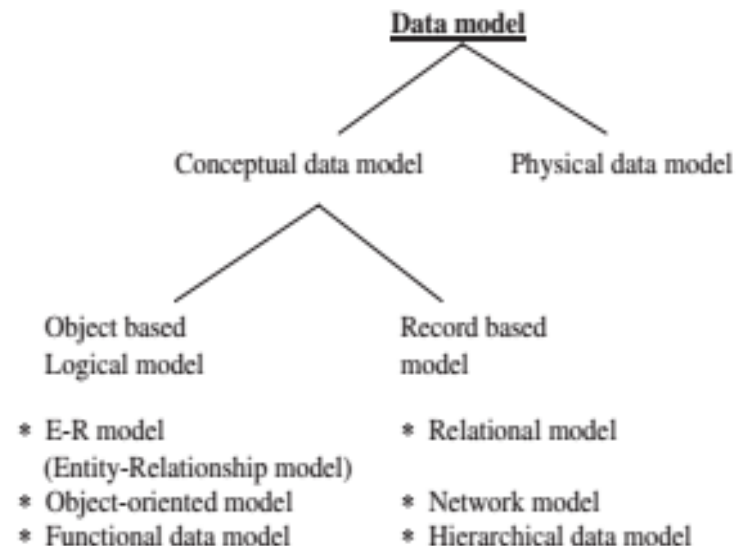
## 2. LEVELS OF DBMS DATA MODELING

Developing a database system involves different levels of abstraction in which we outline the conceptual model and work over steps to form a concrete database schema.

1. Conceptual level.

2. Logical level.

3. Physical level.



## 2. DBMS DATA MODELS

- Data models define how the logical structure of a database is modeled.
- Data Models are fundamental entities to introduce abstraction in a DBMS.
- Data models define how data is connected to each.

## 2.1. CONCEPTUAL LEVEL (Entity-Relationship Model)

- Entity-Relationship (ER) Model is based on the notion of real-world entities(object) and relationships among them.
- While formulating real-world scenario into the database model, the ER Model creates entity set, relationship set, general attributes.
- **Purpose:** The conceptual model aims to capture the business requirements of the system.
- **Output:** Entity-Relationship (ER) diagram

## 2.1.2 Building Blocks of an ER Diagram

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- ✓ ER diagram is a graphical modeling tool to standardize ER modeling.
- ✓ The followings are the basic building blocks of ER diagram:
  - **Entity**
    - Is an object that exists and is distinguishable from other objects, and is uniquely identified.
    - E.g.: A particular person, department, or place
  - **Entity Type**
    - An entity type or entity set is a collection of similar entities
    - E.g.: All students in class, all courses, all departments
  - **Relationship**
    - An association of entities where the association includes one entity from each participating entity type whereas relationship type is a meaningful association between entity types
    - E.g.: *Teaches* is the relationship between LECTURER and STUDENT

## 2.1.2 Building Blocks of an ER Diagram

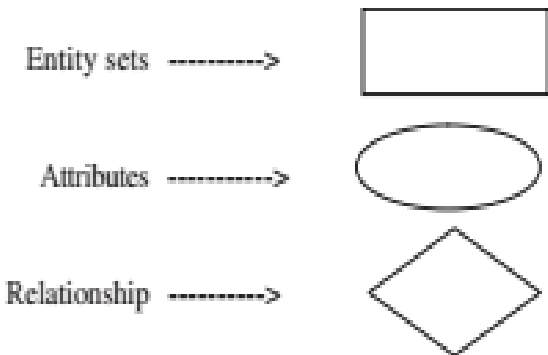
### – Attributes

- Are properties of entity types. In other word, entities are described in a database by a set of attributes.
- E.g.: Student number, names, and grade are the attributes of STUDENT

### ✓ ER Diagram

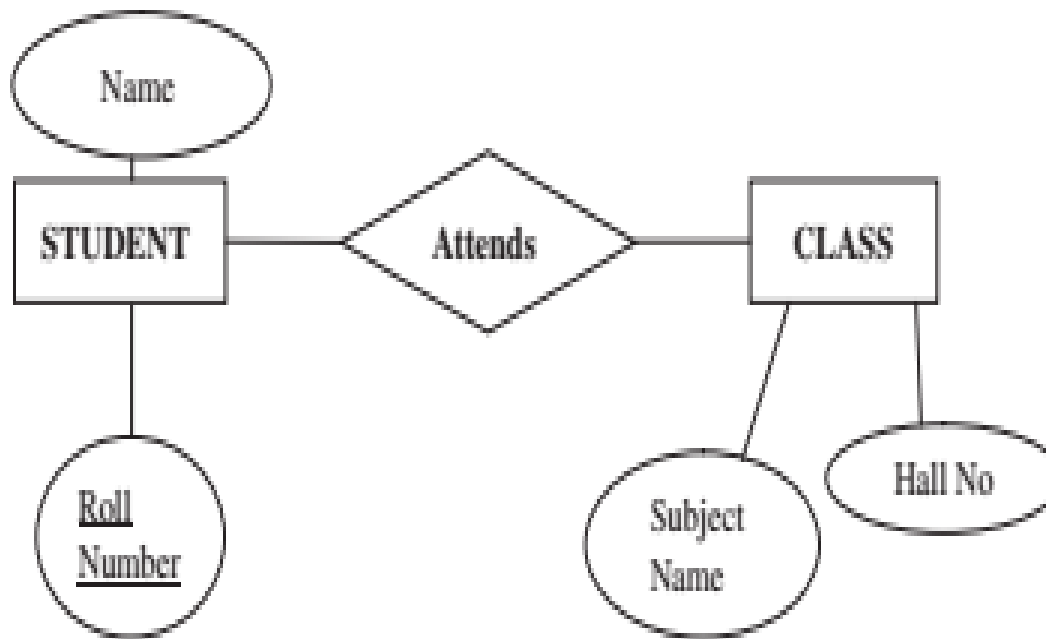
#### – Used to represent database schema

- A rectangle represents an entity set
- An ellipse represents an attribute
- A diamond represents a relationship
- Lines represent linking of attributes to relationship sets



## 2.1.2 Building Blocks of an ER Diagram

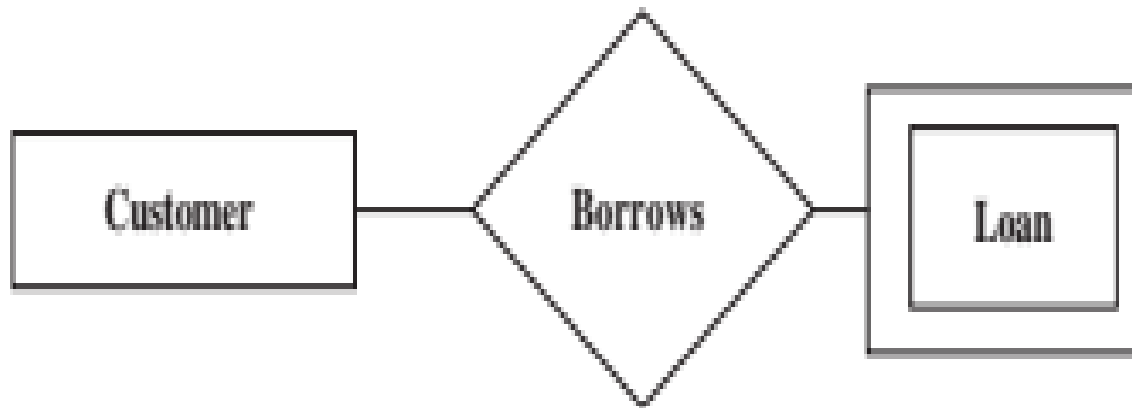
✓ Example of ER diagram



## 2.1.3 Classification of Entity Sets

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- ✓ Entity sets can be broadly classified into:
  - **Strong Entity**
    - Is one whose existence does not depend on other entity
  - **Weak Entity**
    - Is one whose existence depends on other entity.



## 2.1.4 Attribute Classification

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- ✓ An entity is represented by a set of attributes, that is descriptive properties possessed by all members of an entity set.

Example:

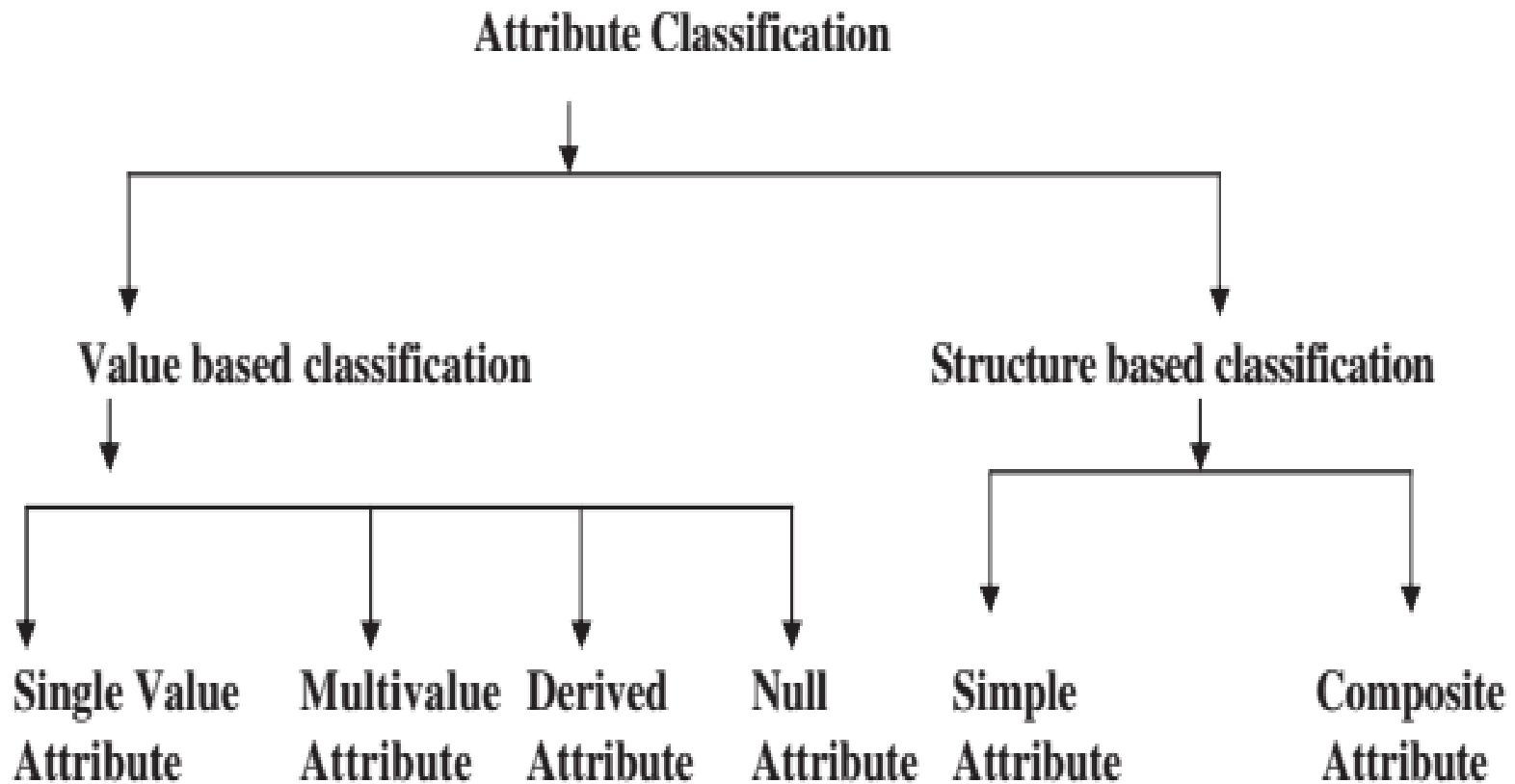
Customer

- customer-id,
- customer-name,
- customer-city

Loan

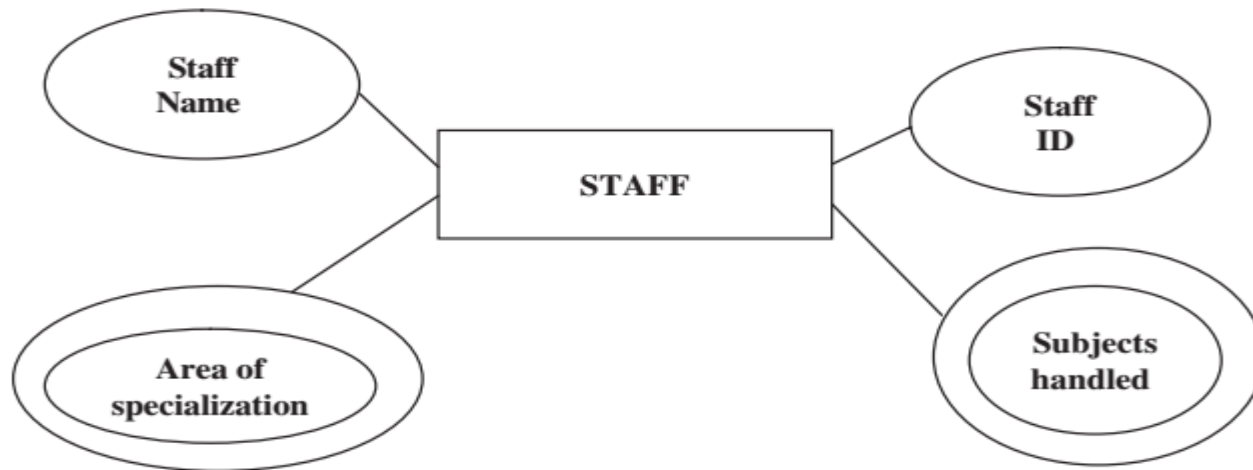
- loan-number,
- amount

## 2.1.4 Attribute Classification



## 2.1.4 Attribute Classification

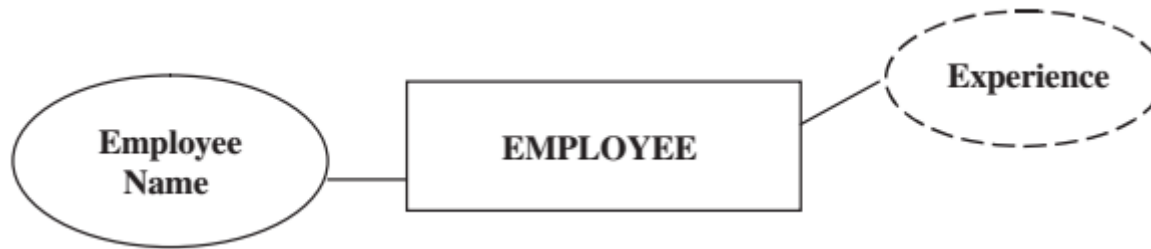
- **Single Value:** There is only one value associated with that attributes
- **Multi-valued:** More than one value will be associated with that attribute
  - Example :multivalued attribute: phone-numbers, Area of specialization



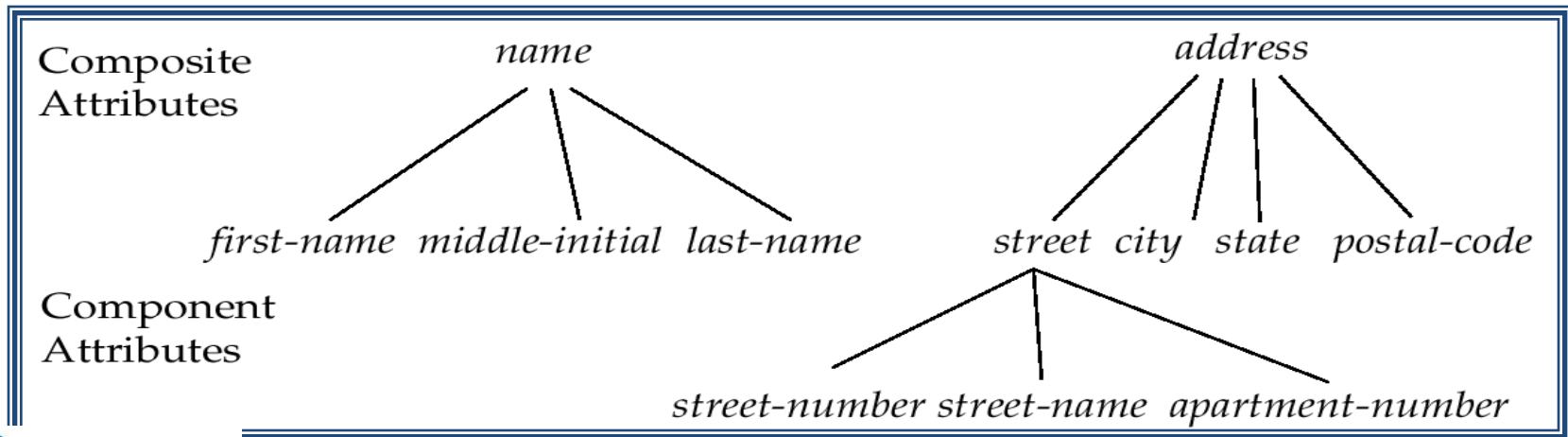
## 2.1.4 Attribute Classification

- ✓ Attribute is used to describe the properties of the entity
  - **Derived:** value of the derived attribute can be derived from the value of other related attributes or entities.

Example: Age.



- **Composite:** attribute which can be further subdivided into simple attributes



## 2.1.4 Attribute Classification

- Symbols used in ER Diagram

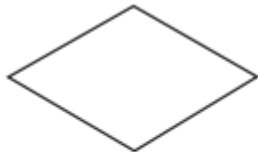
### Basic symbols



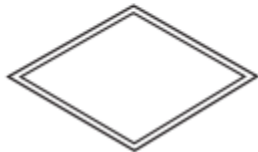
Strong entity



Weak entity



Relationship



Identifying relationship



Associative entity



Attribute



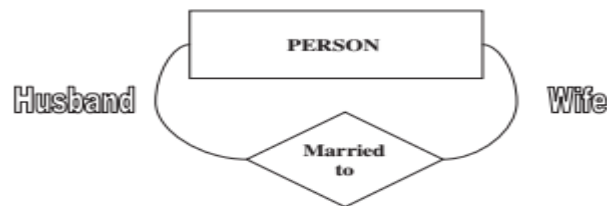
Multivalued attribute



Derived attribute

## 2.1.5 Relationship Degree

- Refers to number of entity sets that participate in a relationship set.
  - Relationship sets that involve two entity sets are binary (or degree two). Generally, most relationship sets in a database system are binary.
  - Relationships between more than two entity sets are rare.
- Unary: also known as recursive relationship. The number of associated entity is one/ entity related to itself.



- Binary: Two entities are involved

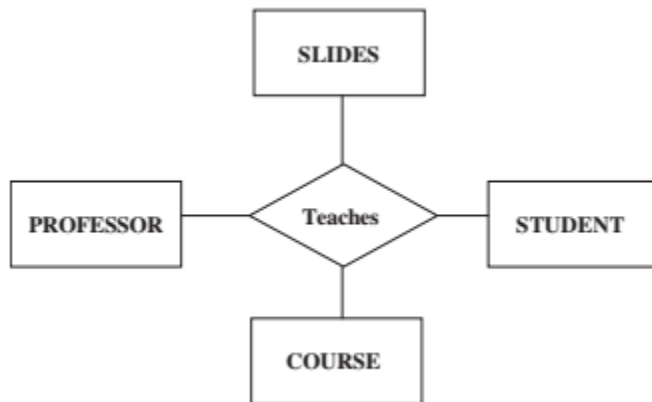


## 2.1.5 Relationship Degree

- ✓ Ternary: Three entities are simultaneously involved

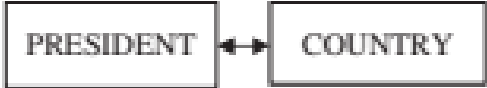
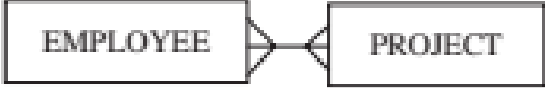
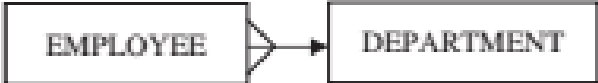


- Quaternary: involve four entities



## 2.1.6 Relationship Classification

- ✓ An association among one or more entities
  - **One-to-Many (1-M)**: associates one entity to more than one entity
  - **One-to-one (1-1)**:
  - **Many-to-Many (N-M)**
  - **Many-to-One (M-1)**

| Relationship type | Representation                                                                      | Example                                                                               |
|-------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| One-to-one        |    |    |
| One-to-many       |    |   |
| Many-to-many      |  |  |
| Many-to-one       |  |  |

# WORK1

## Description :

- You are required to design an Entity-Relationship (ER) diagram for a system that aims to resolve a challenge faced by Rwanda community.
- This could be related to areas such as health, agriculture, education, transportation, environment, or any other sector where you believe technology could offer a solution.

# WORK1

## Steps Involved :

Step 1 : Identify the Entities

Step 2 : Identify the relationships

Step 3: Identify the key attributes

Step 4: Identify other relevant attributes

Step 5: Draw complete ER diagram

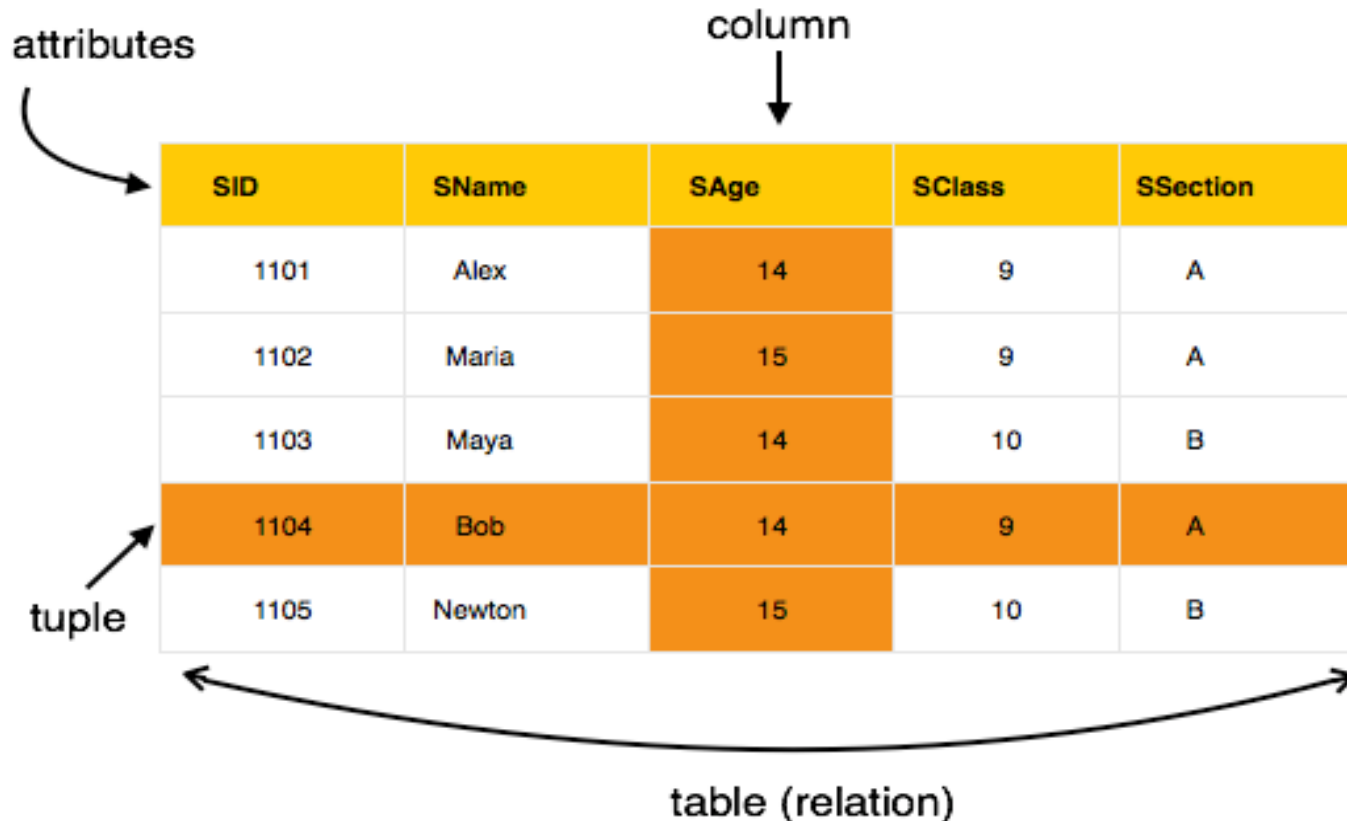


## 2.2.LOGICAL MODELL (RELATION Model)

- The process of creating a Logical model based on a Conceptual model involves:
- The logical level is a more detailed and structured representation, where we focus on how the entities and relationships will be represented in a database.
- But the modeling is still generic as it is not bound to a specific DBMS.
- Output : Relational schema

## 2.2 RELATIONAL MODEL

- The most popular data model in DBMS is the **Relational Model**. It is more scientific a model than others. A **table** is also termed **relation**.



## 2.2.1 RELATIONAL MODEL CONCEPTS

- **Tables** – In relational data model, relations are saved in the format of Tables.
- A table has rows and columns, where **rows represents records (instances)** and **columns represent the attributes**.
- **Tuple** – A single row of a table, which contains a single record for that relation is called a **tuple**.
- **Cardinality of table** – The total number of rows (tuples) in the table.
- **Field** – This is a **column name**, the header which appear at the top of a column. It represents certain characteristics or information about all instances.
- **Degree** – This is number fields (columns) in a table.
- **Cell** – An intersection of a field and a tuple.

## 2.2.1 RELATIONAL MODEL CONCEPTS

- **Relation instance** – A finite set of tuples in the relational database system represents relation instance.
- **Relation schema** – A relation schema describes the relation name (table name), attributes, and their names.
- **Relation key** – Each row has one or more attributes, known as relation key, which can identify the row in the relation (table) uniquely.
- **Attribute domain** – Every attribute has some pre-defined value scope, known as attribute domain.

# 2.2.1 RELATIONAL MODEL CONCEPTS

- Example:

| StudentID | Name  | Age |
|-----------|-------|-----|
| 101       | Alice | 20  |
| 102       | John  | 22  |

- Relation instance :{ (101, Alice, 20), (102, John, 22) }
- Relation schema :This is the structure of the table (table name + columns) :Student(StudentID, Name, Age)
- Relation key :StudentID
- Attribute domain: This is the **allowed values** for a column:
  - StudentID → only integers (e.g., 100–999)
  - Name → only text (e.g., "Alice", "John")
  - Age → integers between 15 and 100

## 2.2.2 Concept of Key

- Key is an attribute or group of attributes, which is used to identify a row in a relation
- Key classification:
  - **Super Key**: A subset of attributes of an entity-set that uniquely identifies the entities/tuples
  - **Candidate Key**: a minimal super key, whose values uniquely identify tuples(rows) in the corresponding relation
  - **Primary Key**: a designed candidate key, must be distinct, and should not be null
  - **Foreign Key**: A set of fields or attributes in one relation that is used to refer to a in another relation

## 2.2.3 CONSTRAINTS

- Every relation has some conditions that must hold for it to be a valid relation.
- There can be any **conditions/restrictions** database designer puts in place to be respected while the database user is entering data in the table.
- These conditions are called **Relational Integrity Constraints**.
- There are three main integrity constraints:
  - **Key constraints**
  - **Domain constraints**
  - **Referential integrity constraints**

# KEY CONSTRAINTS

- There must be at least one minimal subset of attributes in the relation, which can identify a tuple uniquely.
- This minimal subset of attributes is called key for that relation.
- If there are more than one such minimal subsets, these are called candidate keys:
- Key constraints force that:
- In a relation with a key attribute, no two tuples can have identical values for key attributes.
- A key attribute can not have NULL values.

# DOMAIN CONSTRAINTS

- Attributes have specific values in real-world scenario.
- For example:
  - Age can only be a positive integer.
- Every attribute is bound to have a specific range of values.
  - For example,
    - Age cannot be less than zero
    - telephone numbers cannot contain a digit outside 0-9.

StudentID → only integers (e.g., 100–999)

Name → only text (e.g., "Alice", "John")

Age → integers between 15 and 100

# REFERENTIAL INTEGRITY CONSTRAINTS

- Referential integrity constraints work on the concept of Foreign Keys.
- A foreign key is a key attribute of a relation that can be referred in other relation.
- Foreign key column in foreign table depends on base key column in the base table.
- Domain of foreign key column in foreign table is actually set of values which exists in base key column of base table.

## 2.2.4 ER MODEL TO RELATIONAL MODEL

- ER Model, when conceptualized into diagrams, gives a good overview of entity-relationship which is **easier to understand**.
- ER diagrams can be mapped to **relational schema**, that is, it is possible to create **relational schema** using **ER diagram**.

## 2.2.4 Reducing ER diagram to Tables

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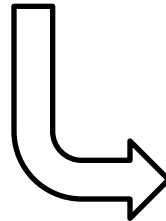
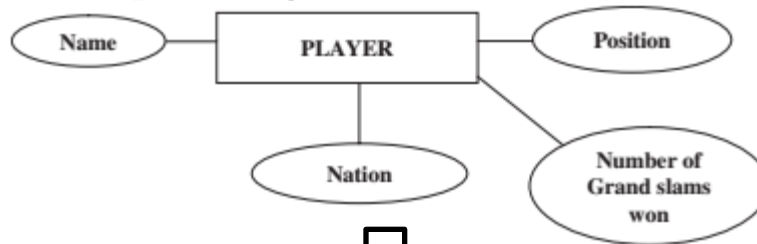
### Mapping Algorithm

- **1 to 1 relationship** with total participation: **Country has president** : The weak entity is ignore and it attributes become the attribute of strong entity. Otherwise make two separate table : example: **User & Profile**
- **1 to many**: **Department and employee**: we create two tables, the strong entity key become a foreign key in the weak entity table.
- **M-N** : **Employee and projects**: The two tables are created + a joint table whereby the primary keys of the two tables become a foreign key in the joint table.

## 2.2.4 Reducing ER diagram to Tables

- Mapping Regular Entities

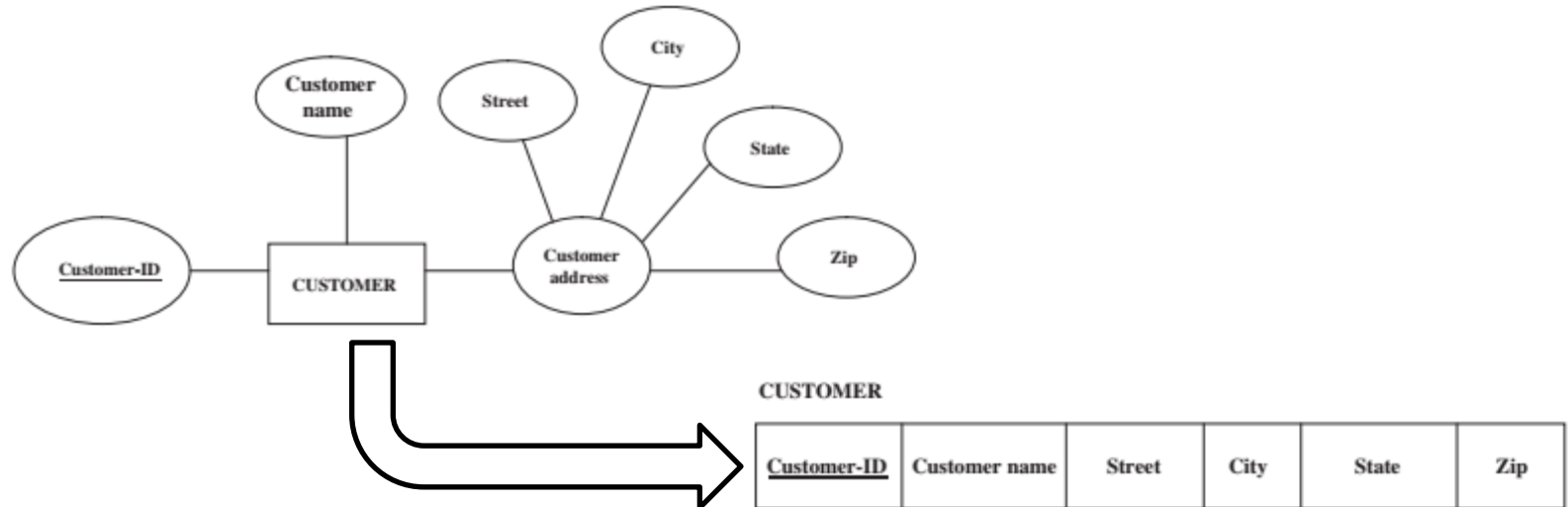
- Each regular entity type in an ER diagram is transformed into a relation. The name given to the relation is generally the same as the entity type
- Each simple attribute of the entity type becomes an attributes of the relation.
- The identifier of the entity type becomes the primary key of the corresponding relation



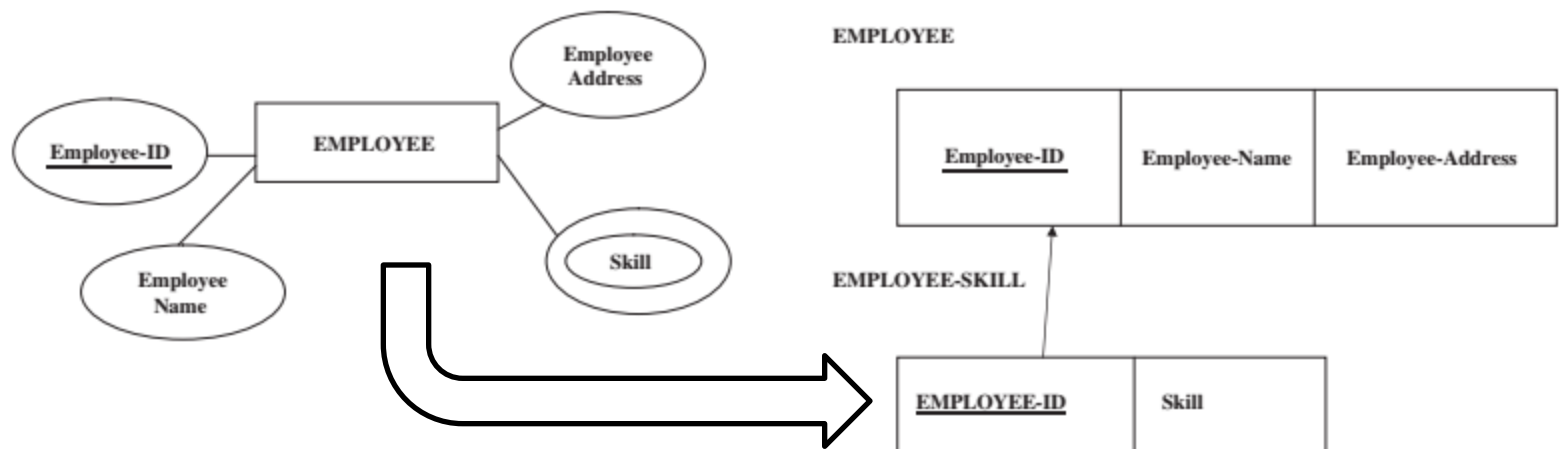
| Player Name   | Nation      | Position | Number of Grand slams won |
|---------------|-------------|----------|---------------------------|
| Roger Federer | Switzerland | 1        | 5                         |
| Roddick       | USA         | 2        | 4                         |

## 2.2.4 Reducing ER diagram to Tables

- Mapping Composite attribute in ER Diagram to tables

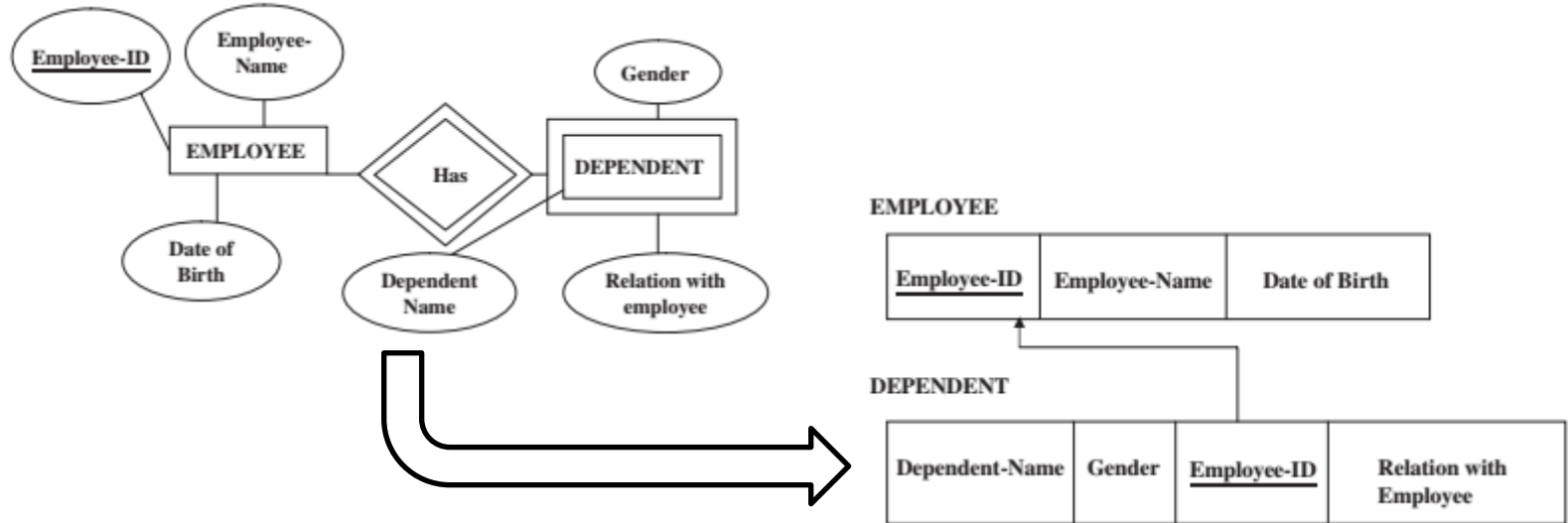


- Mapping Multi-valued attribute in ER Diagram to tables

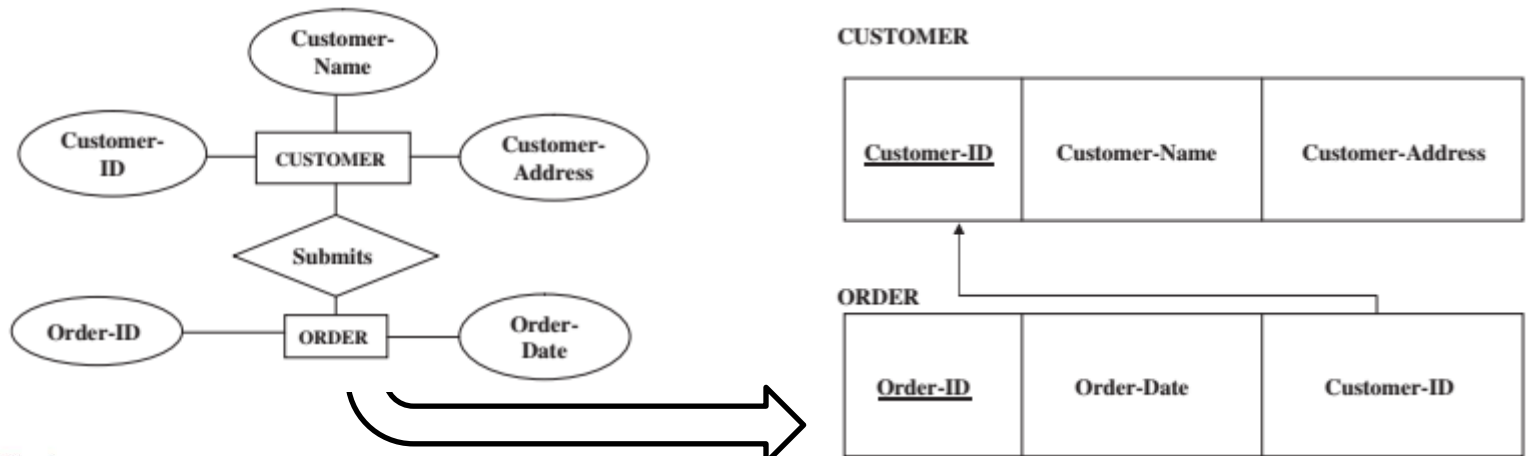


## 2.2.4 Reducing ER diagram to Tables

- Converting Weak Entities in ER diagram to Tables

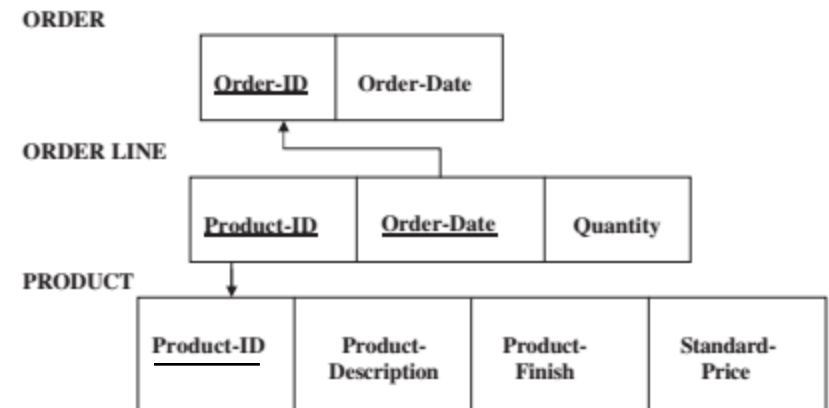
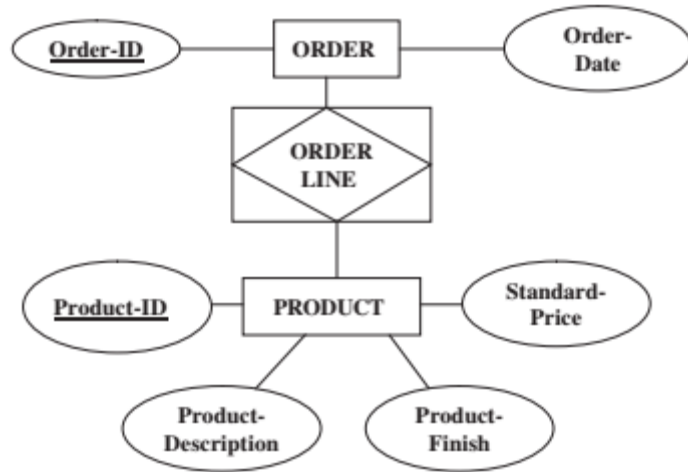


- Converting Binary Relationship to Tables (1-M)

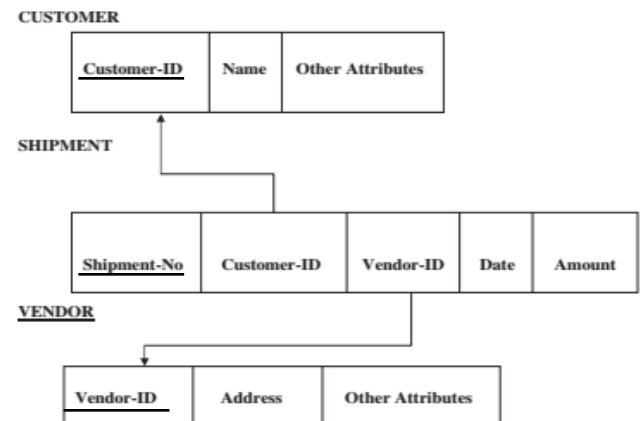
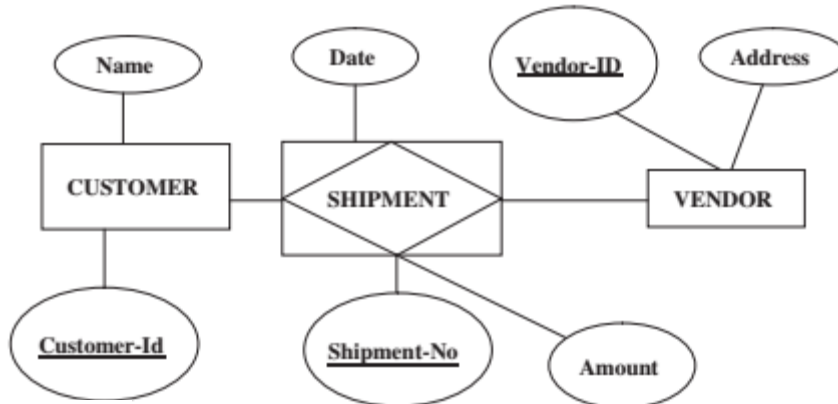


## 2.2.4 Reducing ER diagram to Tables

- Mapping Associative Entity to Tables (N-M) [Without Identifier]

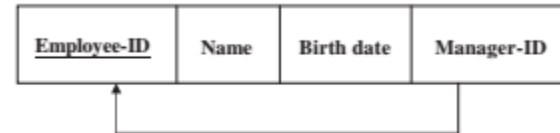
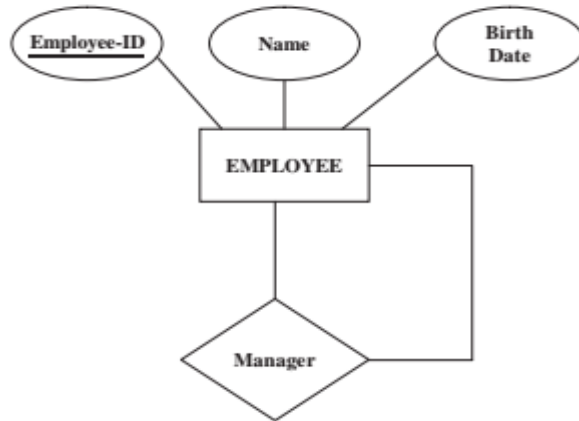


- Mapping Associative Entity to Tables (With Identifier)

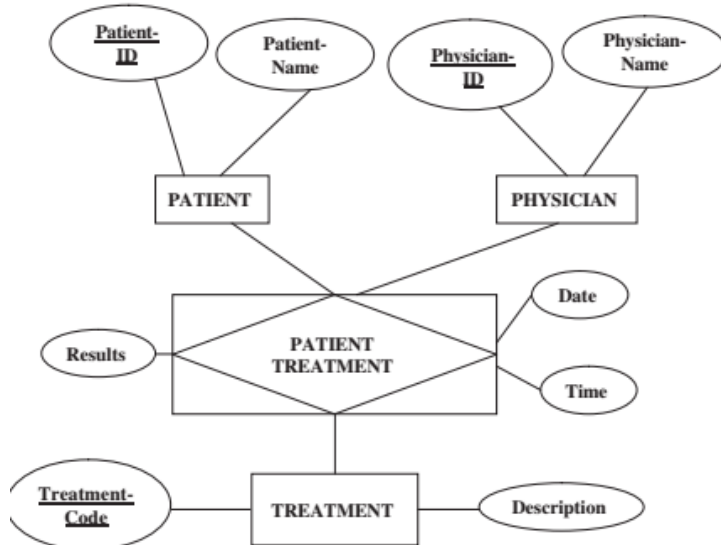


## 2.2.4 Reducing ER diagram to Tables

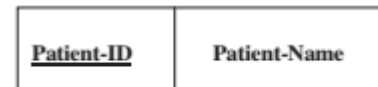
- Converting Unary Relationship to Tables (1-M)



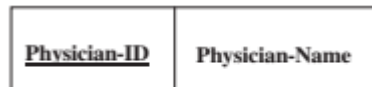
- Converting Ternary Relationship to Tables



PATIENT TREATMENT



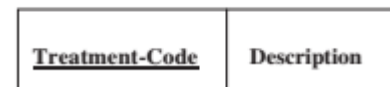
PHYSICIAN



PATIENT TREATMENT



TREATMENT



# Work 2

## Reducing ER diagram to Relational Diagram

# End

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